

§ 2.2.2.

P1. Solution: Characteristic equation

$$r^2 - 6r + 9 = 0$$

$$(r-3) = 0$$

$$r_1 = r_2 = 3.$$

Hence

$$y(t) = C_1 e^{3t} + C_2 t e^{3t}$$

P2. Solution: Characteristic equation

$$4r^2 - 12r + 9 = 0$$

$$(2r-3)^2 = 0$$

$$r_1 = r_2 = \frac{3}{2}$$

Hence

$$y(t) = C_1 e^{\frac{3}{2}t} + C_2 t e^{\frac{3}{2}t}$$

P3. Solution: Characteristic equation

$$9r^2 + 6r + 1 = 0$$

$$(3r+1)^2 = 0$$

$$r_1 = r_2 = -\frac{1}{3}$$

Thus,

$$y(t) = C_1 e^{-\frac{1}{3}t} + C_2 t e^{-\frac{1}{3}t}$$

for some C_1 and C_2 .

$$y(0) = 1 \Rightarrow C_1 = 1$$

$$y(t) = -\frac{1}{3}C_1 e^{-\frac{1}{3}t} + C_2 e^{-\frac{1}{3}t} - \frac{1}{3}C_2 t e^{-\frac{1}{3}t}$$

$$y'(0) = 0 \Rightarrow -\frac{1}{3}C_1 + C_2 = 0$$

$$\Rightarrow C_1 = 1, C_2 = \frac{1}{3}$$

Hence

$$y(t) = e^{-\frac{1}{3}t} + \frac{1}{3}t e^{-\frac{1}{3}t}$$

P4. Solution: Characteristic equation:

$$4r^2 - 4r + 1 = 0$$

$$(2r-1) = 0$$

$$r_1 = r_2 = \frac{1}{2}$$

Thus,

$$y(t) = C_1 e^{\frac{t}{2}} + C_2 t e^{\frac{t}{2}}$$

for some constants C_1 and C_2 .

$$y(0) = 0 \Rightarrow C_1 = 0$$

$$y(t) = \frac{1}{2}C_1 e^{\frac{t}{2}} + C_2 e^{\frac{t}{2}} + \frac{1}{2}C_2 t e^{\frac{t}{2}}$$

$$y'(0) = 3 \Rightarrow \frac{1}{2}C_1 + C_2 = 3 \Rightarrow C_2 = 3.$$

So that

$$y(t) = 3t e^{\frac{t}{2}}$$

P6. Solution: Characteristic equation:

$$r^2 + 2r + 1 = 0$$

$$(r+1)^2 = 0$$

$$r_1 = r_2 = -1$$

Thus,

$$y(t) = C_1 e^{-(t-t_0)} + C_2 (t-t_0) e^{-(t-t_0)}$$

$$= C_1 e^{-(t-2)} + C_2 (t-2) e^{-(t-2)}$$

for some constants C_1 and C_2 .

$$y(2) = 1 \Rightarrow C_1 = 1$$

$$y'(t) = -C_1 e^{-(t-2)} + C_2 e^{-(t-2)} - C_2 (t-2) e^{-(t-2)}$$

$$y'(2) = -1 \Rightarrow -C_1 + C_2 = -1 \Rightarrow C_2 = 0$$

So that

$$y(t) = e^{-(t-2)}$$

P10. Solution: Assume $t > 1$.

$$y_1(t) = t+1, \quad p(t) = \frac{-2(t+1)}{t^2+2t-1}$$

$$u(t) = \frac{e^{-\int p(t) dt}}{y_1^2(t)} = \frac{e^{\int \frac{2(t+1)}{t^2+2t-1} dt}}{(t+1)^2}$$

$$= \frac{e^{\int \frac{2(t+1)}{(t+1)^2-2} dt}}{(t+1)^2} = \frac{e^{\ln((t+1)^2-2)}}{(t+1)^2}$$

$$= \frac{(t+1)^2-2}{(t+1)^2}$$

$$= 1 - \frac{2}{(t+1)^2}$$

$$\begin{aligned}
 y_2(t) &= y_1(t) \int u(t) dt \\
 &= (t+1) \int \left(1 - \frac{2}{(t+1)^2}\right) dt \\
 &= (t+1) \left[t + \frac{2}{t+1}\right] \\
 &= t^2 + t + 2
 \end{aligned}$$

Therefore, $y(t) = C_1(t+1) + C_2(t^2 + t + 2)$

P13. Solution:

$$y_1(t) = t, \quad p(t) = \frac{-2t}{1+t^2}$$

$$\begin{aligned}
 u(t) &= \frac{e^{-\int p(t) dt}}{y_1^2(t)} \\
 &= \frac{e^{\int \frac{2t}{1+t^2} dt}}{t^2} = \frac{e^{\ln(1+t^2)}}{t^2} \\
 &= \frac{1+t^2}{t^2} = 1 + \frac{1}{t^2}
 \end{aligned}$$

$$\begin{aligned}
 y_2(t) &= y_1(t) \int u(t) dt \\
 &= t \int \left(1 + \frac{1}{t^2}\right) dt \\
 &= t \left(t - \frac{1}{t}\right) \\
 &= t^2 - 1
 \end{aligned}$$

Therefore,

$$y(t) = C_1 t + C_2(t^2 - 1)$$